

What Explains Variation in National Cyber Doctrines Across Democracies?

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1 Introduction

In March 2023, the U.S. National Cybersecurity Strategy formally reiterated the Department of Defense’s defend-forward approach, stating that the United States would disrupt and dismantle malicious cyber activity through all instruments of national power (The White House 2023) (U.S. Department of Defense 2023). Germany’s Federal Office for Information Security (BSI) issued public advisories and preventive guidance in the wake of ransomware attacks in critical sectors, including the healthcare industry, in line with the importance of resilience in its 2022 strategy (Bundesamt für Sicherheit in der Informationstechnik (BSI) 2022). India’s activation of the Defence Cyber Agency in 2019 marked a shift toward integrating offensive capabilities into its cyber posture (International Institute for Strategic Studies 2021) (Press Information Bureau, Government of India 2021). These are three democracies, vulnerable to common cybersecurity threats, with increasingly different cybersecurity policies.

This divergence creates urgent problems for alliance coordination. NATO’s cyber defense framework assumes doctrinal harmony between democratic nations within the organization (North Atlantic Treaty Organization 2016a) (North Atlantic Treaty Organization 2023), and yet the member nations diverge on what constitutes an attack that calls for retaliation instead of regulation. Can alliances be effective deterrents if nations in those alliances have different threat perceptions and retaliation abilities?

This essay asserts that cyber doctrine results from three interconnected factors: external threat level, domestic institutional structures regarding privacy and regulation, and alliance relationships, particularly intelligence-sharing partnerships. These factors interact rather than operate independently. Domestic institutions, including constitutional privacy protections and independent regulatory agencies, filter international threats and shape states' offensive versus defensive orientations.

Through a structured comparison of the United States (offense-dominant), Germany (defense-dominant), and India (hybrid/emerging offensive), this paper demonstrates that institutional constraints exert the most substantial independent influence on cyber doctrine. While threat environment matters and alliances create coordination pressures, democracies with robust privacy protections maintain defensive doctrines even under severe threat. Conversely, democracies with weak institutional constraints allow threat pressure to translate directly into offensive capability development.

These findings have immediate policy implications. First, NATO's pursuit of doctrinal convergence overlooks the institutional foundations of divergence; Germany is unlikely to adopt U.S.-style operations without constitutional change. Secondly, deterrence in joint action is compromised by institutional diversity. Thirdly, any assistance to emerging democracies like India in building cyber-capabilities should prioritize institutional safeguards in order to maintain those abilities within democratic parameters.

2 Theoretical Framework: Why Democracies Choose Different Cyber Doctrines

2.1 The Institutional Foundations of Military Doctrine

Barry Posen established that military doctrine emerges from interactions between international threats, domestic political structures, and civil-military relations (Posen 1984). Peter Katzenstein's comparative analysis of German and Japanese counterterrorism further demonstrates how legal and normative systems and political histories generate systematic policy divergence among democratic allies facing comparable threats (Katzenstein 2003). This institutional logic applies di-

rectly to cyber policy: just as privacy constraints shape counterterrorism, they also structure cyber operations involving surveillance, attribution, and intelligence collection.

The cybersecurity literature remains fragmented on doctrinal determinants. Joseph Nye argues that classical deterrence concepts apply to cyberspace but require modification due to attribution difficulties and the offense–defense balance (Nye 2017). Jon Lindsay shows that organizational capacity, not merely technical capability, determines whether and how states can integrate cyber operations into larger strategic frameworks (Lindsay 2014). Yet much existing work treats democracies as a homogeneous category, without examining the significant variation within democracies themselves (Kostyuk and Zhukov 2019) (Libicki 2009). This paper addresses that gap by foregrounding institutional diversity among democracies.

2.2 Defining Cyber Doctrine

Following Posen’s conceptualization of doctrine, cyber doctrine refers to the principles, structures, and operational concepts that guide a state’s use of cyber capabilities to achieve national security goals. Doctrines vary along an offensive–defensive continuum:

Offensive doctrines emphasize pre-emptive or proactive cyber operations (“defend forward”), legal frameworks enabling operations abroad, standing forces for offensive action (e.g., U.S. Cyber Command), and budgetary prioritization of offensive capability development (U.S. Congress 2018) (U.S. Department of Defense 2023).

Defensive doctrines prioritize resilience, incident response, critical-infrastructure protection, regulatory frameworks, and civilian cybersecurity agencies with defensive mandates (Federal Ministry of the Interior and Community 2021) (Bundesamt für Sicherheit in der Informationstechnik (BSI) 2022)

Most democratic systems are mixed in orientation, and state coding can be done in accordance with their predominant stress in strategic literature, institutional setup, spending, and declared activities.

2.3 Hypotheses and Causal Mechanisms:

2.3.1 H1: Threat Environment

Countries that experience frequent, intense cyberattacks may adopt an offense-dominant strategy. Continuous attacks create political motivation for action and reveal the weaknesses of purely responsive action. In such situations, the states involved may conclude that something more proactive or preventive is needed. This is consistent with the logic that when threats intensify or when actors perceive an advantage in offense, states become more willing to adopt preemptive or preventive strategies. (Van Evera 1999) Threats are understood through institutional perspectives; a single attack can result in a greater focus on resilient capabilities for a democracy or a responsive attack for another.

Observable implication: Democracies facing frequent cyberattacks from state adversaries, especially where cyber operations intersect with military disputes, should develop offensive capabilities.

Falsification: If high-threat democracies consistently maintain defensive doctrines despite sustained attacks, H1 is insufficient.

2.3.2 H2: Institutional Structures

Democracies with strong constitutional privacy protections and autonomous regulatory agencies are less likely to develop and use offensive cyber doctrines. Norms and legal frameworks are essential because they impose political and judicial costs on governments to carry out any type of surveillance (Katzenstein 2003). Regulatory bodies such as the German BSI develop bureaucratic interests aligned with defensive approaches (Federal Ministry of the Interior and Community 2021). Constitutional privacy guarantees (Federal Republic of Germany 1949) create hard legal limits on surveillance-dependent offensive cyber operations.

Institutional effects depend heavily on the distinction between domestic and foreign operations. For instance, the United States offers strong privacy laws to its citizens, but also conducts

offensive activities targeting foreign targets.

Observable implication: Even when facing serious cyber threats, democracies that enjoy constitutional privacy safeguards and strong regulatory bodies should retain defensive postures.

Falsification: If such democracies rapidly develop offensive capabilities in response to attacks despite intact privacy safeguards, H2 is refuted.

2.3.3 H3: Alliance Commitments

Highly institutionalized intelligence-sharing alliances (e.g., Five Eyes) increase pressure for doctrinal convergence due to increased burden-sharing and threat perceptions (Gioe, Goodman, and Schaefer 2021). However, this is not the case in each of the alliances in question. NATO member states, for instance, are well along in their level of cyber collaboration, and nation-states have the final say in what happens on the ground in terms of increased doctrinal variation (North Atlantic Treaty Organization 2023).

Observable implication: Members of deeply integrated alliances are expected to be more aligned in terms of their doctrines, whereas less integrated alliances are expected to display larger variation.

Falsification: If deeply integrated allies exhibit more divergence than loosely integrated states despite institutional integration, H3 requires refinement.

2.4 Interaction Effects

These hypotheses generate cross-pressures. A democracy facing frequent intrusions (H1 → offense) but possessing strong privacy protections (H2 → defense) must reconcile conflicting incentives. The outcome depends on: (1) the salience and severity of the threat; (2) the strength and autonomy of regulatory and judicial institutions; and (3) whether elites can frame offensive operations as compatible with democratic constraints.

Alliance pressures (H3) may amplify or dampen these tensions. Deeply institutionalized alliances may encourage offensive capability development, while democracies with strong privacy

frameworks may resist such pressures by maintaining defensive postures aligned with domestic legal constraints.

3 Methodology and Case Selection

This paper employs a structured, focused comparison of three democratic cases to evaluate competing hypotheses (George and Bennett 2005). The approach allows process tracing to identify causal mechanisms (Bennett and Checkel 2015) while enabling controlled comparison across similar regime types. Cases were chosen to capture variation in cyber doctrine while keeping regime type constant in order to distinguish between institutional and strategic variation within democratic regimes.

3.1 United States: Offense-Dominant:

The 2018 National Defense Authorization Act authorized U.S. Cyber Command to conduct persistent engagement abroad, establishing the defend-forward approach (U.S. Congress 2018). The 2023 DoD Cyber Strategy reiterated this posture (U.S. Department of Defense 2023). The United States faces high-intensity cyber threats (H1), serves as the core of the highly institutionalized Five Eyes alliance (contra H3) (Gioe, Goodman, and Schaefer 2021), and operates within privacy-protective domestic legal frameworks (H2). This combination makes the U.S. an ideal case for testing how severe threats interact with moderate institutional constraints.

3.2 Germany: Defense-Dominant:

Germany's Cybersecurity Strategy 2021 emphasizes digital sovereignty, regulation, and resilience rather than offensive operations (Federal Ministry of the Interior and Community 2021). Despite significant intrusions, most notably the 2015 Bundestag hack attributed to Russian APT28 (Bundesamt für Sicherheit in der Informationstechnik 2015), Germany strengthened its defensive institutions rather than developing offensive capabilities. Strong constitutional privacy protections in the Grundgesetz and the BSI's defensive mandate create structural constraints (H2) that override

both threat pressure (H1) and NATO's increasingly assertive cyber posture (H3) (North Atlantic Treaty Organization 2023).

3.3 India: Hybrid / Emerging Offensive

India faces intense regional cyber threats from Pakistan-linked groups and Chinese state-linked intrusion sets targeting critical infrastructure (Recorded Future Insikt Group 2021). India lacks participation in deeply institutionalized intelligence alliances (contra H3), engaging instead in loose partnerships such as the Quad (Quad Leaders 2021). Weak privacy protections and an incomplete data-protection regime (Ministry of Electronics and Information Technology (MeitY) 2021) reduce institutional constraints (contra H2). The activation of the Defence Cyber Agency (International Institute for Strategic Studies 2021) (Press Information Bureau, Government of India 2021) reflects the translation of threat pressure directly into offensive capability development.

Across the three cases, observable cyber doctrine reflects the interaction of threat severity, institutional constraints, and alliance structures, with domestic institutions emerging as the most consistent mediating factor.

4 Case Study

4.1 United States: Offense-Dominant Doctrine

The United States represents the most transparent case of an offense-dominant democratic cyber doctrine. In 2018, the National Defense Authorization Act expanded Cyber Command's role. It highlighted its mandate for persistent engagement with foreign hostile actors and for improving the approval process for select lower-than-threshold activities (U.S. Congress 2018). In 2023, the National Cybersecurity Strategy reaffirmed this framework and committed to leveraging the full range of national power to disrupt foreign malignant cyber activities (The White House 2023). Several instances of publicly disclosed US cyber actions would further point to the ongoing implementation of an offensive approach, such as hacking attacks on IS-related targets (Sanger 2016) or

on targets aimed at undermining Russia during the 2018 midterm election ((Nakashima 2018).

4.1.1 Explaining U.S. Offensive Doctrine

I. Testing H1: Threat Environment

The U.S. case strongly supports H1. The United States is under persistent, sophisticated cyberattacks launched by several state-level actors. China-based state actors have been implicated in espionage attacks on defense contractors and government systems (International Institute for Strategic Studies 2021). Russia's attacks include election hacking, attacks on critical infrastructure, and attacks on federal government systems (NSA, CISA, FBI 2021). Iran has perpetrated several destructive cyber attacks, including the attacks on Saudi Arabia's Saudi Aramco (Alghamdi 2021), Operation Ababil (EBSCO Research Starters 2021), and those on Las Vegas Sands in 2014 (Reuters 2014). North Korea carried out a cyber attack on Sony Pictures that demonstrated its political retaliation capacity (Federal Bureau of Investigation 2014).

Large-scale incidents include the incident in 2015 at the US Office of Personnel Management, in which a massive number of records were compromised (U.S. House Committee on Oversight and Government Reform 2016), and the Solar Winds incident in 2020. Joint alerts in 2020 and 2021 drew attention to the large-scale nature of incidents such as SUNBURST and implied that only a purely defensive approach is no longer sufficient.

Process tracing aligns with H1: the rise in foreign intrusion activity pushed top US officials, such as past NSA and CYBERCOM leaders, to state that authorities were hindering efforts to act in time. Then, the clarification of CYBERCOM's authorities in the 2018 NDAA enabled steady engagement. Later strategy statements codified this approach (U.S. Department of Defense 2023), signifying a threat-based change to an offensive philosophy.

II. Testing H2: Institutional Structures

Reforms to surveillance law and the FISA Court oversight in the wake of Snowden include strong safeguards against domestic surveillance (United States Congress 2015) (United States Congress 1978). However, the legal system differentiates between domestic and foreign surveil-

lance. Foreign intelligence collection is governed by more permissive authorities, allowing offensive cyber operations abroad within established oversight channels. CISA coordinates civilian cyber defense, but even as it coordinates the country's cyber defenses, it lacks the powers of Germany's BSI to impose technical standards on this infrastructure (Cybersecurity and Infrastructure Security Agency 2025) (Bundesamt für Sicherheit in der Informationstechnik (BSI) 2023). Thus, some factors that can limit the scope of cyber operations of the United States include institutional constraints within the country. H2 therefore moderates but does not impede offensive doctrine.

III. Testing H3: Alliance Structures

The U.S. occupies the leading position in deeply institutionalized intelligence alliances such as Five Eyes, which generate shared intelligence architectures and common operational expectations (Gioe, Goodman, and Schaefer 2021). NATO's maturing cyber posture, increasingly incorporating offensive elements at the declaratory level, reinforces these dynamics (North Atlantic Treaty Organization 2023). Because many allies face legal and political barriers to developing offensive capabilities, U.S. leadership in cyber offense can function as a form of alliance burden-sharing.

The U.S. case provides strong confirmation for H1 and partial support for H3. Institutional constraints (H2) function as boundary conditions, shaping but not preventing an offense-dominant doctrine enabled by high threat intensity and alliance responsibilities.

4.2 Germany: Defense-Dominant Doctrine

Germany's cyber posture is shaped far more by legal institutional constraints than by threat severity or alliance pressure. The 2021 Cybersecurity Strategy stresses digital sovereignty, protection of critical infrastructure, and resilience, and eschews any type of cyber operation that is offensive (Federal Ministry of the Interior and Community 2021). In Germany, the leading organization for cybersecurity is the BSI, a purely civilian institution that exclusively concentrates on standards, certification of technologies, warnings, and incidents (Bundesamt für Sicherheit in der Informationstechnik (BSI) 2022). It lacks authority for intrusive collection or offensive activity, and Ger-

many has no equivalent to U.S. Cyber Command.

The Bundeswehr's Cyber and Information Domain Service (KdoCIR), established in 2017 (Bundeswehr 2017), operates under tight parliamentary oversight and constitutional proportionality rules, which significantly constrain any potential for offensive action. Germany's institutional design, therefore, structurally favors defense (Federal Constitutional Court of Germany 2008) (Federal Constitutional Court of Germany 2020).

4.2.1 Explaining Germany's Defensive Doctrine

I. Testing H1: Threat Environment Germany faces persistent, high-level cyber threats. The 2015 Bundestag hacking incident affected the country's legislature for several months (Bundesamt für Sicherheit in der Informationstechnik 2015). Annual reports from BfV reveal that the country's industries are continually spied on by both Chinese and Russian operators (Bundesamt für Verfassungsschutz (BfV) 2022).

Under H1, this threat intensity should generate pressure for offensive capability development. Yet Germany reacted in the opposite direction. Rather than authorizing retaliation or building offensive tools, policymakers expanded BSI competencies and updated the IT Security Acts to mandate stronger defensive standards. Public discourse focused on resilience, attribution, and EU-level regulation, not offensive cyber power. This makes Germany strong contradictory evidence for H1.

II. Testing H2: Institutional Structures

Germany's constitutional order provides a straightforward explanation for this defensive orientation. Article 10 of the Grundgesetz establishes strong guarantees of privacy, which have been continually expanded to cover digital communication (Federal Republic of Germany 1949). In two landmark decisions in 2008 and 2020, the Court declared large-scale surveillance and foreign intelligence laws unconstitutional, asserting that internationally routed communications are also protected by the constitution and must satisfy very stringent standards of proportionality (Federal Constitutional Court of Germany 2008) (Federal Constitutional Court of Germany 2020).

They present tough legal hurdles for offensive cyber attacks that need intrusive intelligence collection or access to the network. Policymakers anticipate judicial rejection, leading to political reluctance to authorize offensive structures.

Process-tracing: After the Bundestag hack, debate briefly opened around more assertive capabilities, but the 2020 BVerfG ruling sharply constrained foreign surveillance, effectively ending momentum for offensive expansion and pushing investment toward the BSI and defensive EU-wide regulation. This sequence, *constitutional limits* → *judicial rulings* → *preemptive political caution* → *defensive reinforcement*, aligns directly with H2's causal mechanism.

III. Testing H3: Alliance Structures

Germany participates actively in NATO's cyber decision-making, including in the 2016 Warsaw Declaration, which states that cyberattacks may invoke Article 5 (North Atlantic Treaty Organization 2016b). However, its approach to cyber conflict emphasizes defense and protection of privacy, and it seeks to extend those values in the EU (GDPR, NIS Directive) (European Union 2016b) (European Union 2016a) by insisting on defensive approaches within alliances rather than aligning with US or UK views that favor offense.

Thus, alliance pressure does not override domestic institutions; rather, it is filtered through them. Germany relies on alliances to promote its defensive orientation rather than adopt more offensive strategies.

Germany strongly supports H2 and contradicts H1. Severe threats did not translate into an offensive doctrine because constitutional privacy protections, judicial oversight, and the BSI's institutional design impose binding constraints. H3 holds only in the mediated sense: Germany shapes alliance approaches rather than adopting offensive norms. Germany demonstrates how domestic institutions can anchor a defense-dominant doctrine even under persistent external threats.

4.3 India: Hybrid Doctrine and Emerging Offensive Capabilities

India represents a hybrid doctrinal trajectory: historically defensive, but increasingly oriented toward offensive cyber capabilities. This evolution results from a severe regional threat environment (H1) operating in a context of comparatively weak institutional constraints (H2) and limited alliance integration (H3). Unlike the United States or Germany, India's cyber policy has developed with substantial autonomy from external doctrinal pressures and with minimal judicial or regulatory limitations in cybersecurity policies.

Initially, the Indian cybersecurity framework remained very much on the defensive side. The incident response entity, Computer Emergency Response Team (CERT-In), was formed in 2004 (Indian Computer Emergency Response Team (CERT-In) 2004). In 2014, the National Critical Information Infrastructure Protection Centre (NCIIPC) emerged, taking on a very narrow protection role in sectors such as energy, finance, transport, and communication (National Critical Information Infrastructure Protection Centre (NCIIPC) 2014). These institutions mirror the defensive-first posture typical of states in early cyber development phases.

However, a paradigm shift emerged with the creation and activation of the Defence Cyber Agency (DCA), integrating cyber functions in the different branches of the forces and emphasizing both offensive and defensive functions (International Institute for Strategic Studies 2021) (Press Information Bureau, Government of India 2021). Suggestions of active cyber operation and indigenous development of offensive cyber capability emerge in declarations made at higher levels in the Defence Ministry.

4.3.1 Explaining India's Hybrid Doctrine

I. Testing H1: Threat Environment

India faces one of the most intense geopolitical threat environments among democracies studied here. Annual reports of CERT-In have highlighted an increased number of incidents of defacement attacks, phishing, and malware attacks happening during or around periods of tension or conflict between certain Indian neighbors (Indian Computer Emergency Response Team (CERT-

In) 2021). Though the Government of India itself has never made any official accusations of such activities, the nature and timing of such activities have been observed to consistently align when it comes to incidents of tension or conflict between India and Pakistan, especially around Kashmir.

However, China poses the most advanced level of cyber threat. Chinese-linked advanced persistent threat (APT) actors have launched attacks on Indian government ministries, the Indian defense sector, maritime networks, and strategic infrastructure. In 2021, Recorded Future reported state-sponsored actors, RedEcho, carrying out intrusion attacks on power infrastructure networks that were connected to load dispatching facilities, which may have been attributed to coercive communication due to border tensions (Recorded Future Insikt Group 2021).

These intrusions occur in the context of intermittent territorial disputes and face-offs along the Line of Actual Control (LAC) between nations. Cyber activities in India are closely connected to the threat of kinetic escalation, unlike Germany, which is threatened purely by espionage activities. Under H1, such acute adversarial pressure should drive the development of offensive cyber capabilities. India's trajectory strongly confirms this expectation: threat severity appears to be the primary driver of doctrinal evolution.

II. Testing H2: Institutional Structures

Institutional constraints in India differ sharply from the German or U.S. cases. Although the landmark Supreme Court decision in Puttaswamy (2017) declared the right to privacy as a fundamental right, it is unfortunate that no binding data protection legislation exists in the country. No effective regime has been established in the country since 2019, despite several attempts at legislation (Supreme Court Observer 2017) (Ministry of Electronics and Information Technology (MeitY) 2021). The Digital Personal Data Protection Act (2023) (Government of India 2023) can be seen as moving in the right direction towards forming a comprehensive regime of privacy, yet in the current early stages of implementation and in light of exemptions granted at the state level, the privacy standards in this country are still remarkably low. As a result, constitutional privacy protections remain doctrinal rather than operational, with limited institutional enforcement capacity.

Regulatory bodies possess constrained authority. CERT-In coordinates incident response but cannot impose binding cybersecurity standards beyond specific reporting rules (Indian Computer Emergency Response Team (CERT-In) 2022b). NCIIPC has a narrow sectoral mandate and limited enforcement power (National Critical Information Infrastructure Protection Centre (NCI-IPC) 2014). India has no equivalent to Germany's BSI or the U.S. CISA, no civilian authority with strong autonomy, standard-setting authority, or institutional incentives to prioritize defensive doctrine.

This institutional weakness has two effects consistent with H2's reverse logic: 1. Few legal barriers exist to expanding offensive military cyber units; 2. Regulatory institutions cannot shape doctrine or mobilize political opposition to offensive development.

Thus, unlike Germany, where courts, privacy norms, and regulatory agencies suppress offensive doctrine, India's institutional environment is permissive. H2 therefore plays a crucial enabling role by not constraining H1-driven doctrinal evolution.

III. Testing H3: Alliance Structures

India's alliance environment provides a valuable negative case for H3. India is not a member of highly institutionalized sharing agreements like the Five Eyes or any NATO alliances. Though it engages in dialogues on issues of cybersecurity with many nations, this leads to very little operational integration and no doctrinally integrated relationships (Ministry of External Affairs (MEA) 2018).

India is also involved in the Quadrilateral Security Dialogue (QUAD), in which the 2021 Joint Leaders' Statement supports collaboration on digital standards and critical technologies, but it does not include intelligence sharing or joint offensive operations (Quad Leaders 2021).

India's offensive development, therefore, occurs largely independently of alliance pressure. This challenges strong versions of H3 by demonstrating that offensive doctrine can emerge without integration into alliance structures, when threat pressures (H1) are strong and institutional constraints (H2) are weak.

India provides the evidence for interaction effects between H1 and H2. Severe, multi-

directional cyber threats from Pakistan and China create strong incentives for offensive capability development, and weak institutional constraints enable these incentives to translate directly into doctrine. Limited alliance integration (H3) does not inhibit offensive development but reinforces India's need for indigenous capabilities. India thus exemplifies a hybrid doctrine shaped primarily by threat intensity and institutional permissiveness rather than alliance pressures.

5 Comparative Analysis and Findings

5.1 Hypothesis Testing

5.1.1 H1: Threat Environment

The U.S. and India provide strong support for H1. Both face sustained, geopolitically embedded cyber threats linked to broader military rivalries. U.S. exposure to Chinese espionage, Russian political interference, and high-impact intrusions such as SolarWinds contributed directly to the adoption of persistent engagement and defend-forward operations. In India, threat escalation from Pakistan-linked actors and Chinese state-linked groups targeting critical infrastructure, most notably RedEcho's 2021 intrusions into Indian power systems, correlates with the rapid expansion of offensive cyber capability following the activation of the Defence Cyber Agency (International Institute for Strategic Studies 2021) (Press Information Bureau, Government of India 2021).

In contrast, Germany offers strong disconfirming evidence. Despite the very intrusive attacks, such as the Bundestag hack of 2015, which involved Russian actors of the APT28 group (BSI 2015), Germany never engaged in any kind of offense-related development. Instead, the Bundestag hack had legal, regulatory, and organizational implications that were mainly focused on resilience.

Refinement: H1 holds only when (1) cyber threats are connected to plausible military contingencies, and (2) domestic institutional structures allow governments to pursue offensive responses. In democracies facing espionage-oriented threats without strong territorial or military

drivers (Germany), threat severity does not translate into offensive doctrine.

5.1.2 H2: Institutional Structures

H2 receives the strongest and most consistent support across all cases. Germany’s constitutional guarantees (Article 10) (Federal Republic of Germany 1949) striking down far-reaching surveillance powers (Federal Constitutional Court of Germany 2008) (Federal Constitutional Court of Germany 2020) impose constitutional limitations on aggressive surveillance and, procedurally, also on hacking attacks. Additionally, the administrative bodies of the country, in particular the BSI, maintain the role of the defender in favor of resilience.

The United States exhibits moderate constraints: strong domestic privacy protections coexist with permissive foreign-targeting authorities under FISA and related frameworks (United States Congress 1978) (United States Congress 2015). This institutional asymmetry allows an offensive doctrine focused on external operations while generating political contestation over domestic surveillance. The U.S. demonstrates that institutional constraints shape where offense occurs, not if it occurs.

India represents the opposite end of the institutional spectrum. Though the Supreme Court of India in Puttaswamy (2017) holds the right to privacy as a basic right (Supreme Court Observer 2017), it still doesn’t have full-fledged data protection laws (Ministry of Electronics and Information Technology (MeitY) 2021). Regulatory authorities, CERT-In (Indian Computer Emergency Response Team (CERT-In) 2022a) and NCIIPC (National Critical Information Infrastructure Protection Centre (NCIIPC) 2014), have coordination powers but hardly any independent powers or standardization abilities. Because of this, threat pressures lead to advancements in offensive doctrines largely unhindered.

Across cases, a clear gradient emerges: *strong institutions* → *defensive doctrine (Germany)*; *moderate institutions* → *offense with constraints (U.S.)*; *weak institutions* → *rapid, unconstrained offensive development (India)*.

5.1.3 H3: Alliance Structures

H3 receives mixed support. Alliance membership alone cannot explain doctrinal outcomes. Both the US and Germany are members of NATO, yet each maintains a different cyber doctrine. India inclines towards an offensive cyber doctrine despite its low level of integration into alliances, and this indicates that for some nations, joining formal alliances may neither be necessary nor sufficient for the pursuit of offensive doctrine.

Alliance effects depend on: 1. Institutional depth: Five Eyes' highly institutionalized intelligence architecture creates stronger convergence pressures than NATO's looser coordination (Gioe, Goodman, and Schaefer 2021).

2. Power dynamics :dominant actors like the U.S. shape alliance standards rather than being constrained by them.

3. Domestic institutional mediation: Germany resists offensive alignment due to constitutional limits; the U.S. accommodates alliance expectations because domestic law permits foreign-focused offensive operations.

Refinement: Alliances influence doctrine only when deep institutionalization combines with permissive domestic institutions. Where institutions are restrictive, states reinterpret alliance commitments through a defensive lens.

5.2 Alternative Explanations Assessed

Technological capacity does not predict doctrinal outcomes. Germany remains a global leader in cyber technology while upholding its defensive doctrine. India's massive IT workforce, over 5 million workers (NASSCOM 2023), does not translate automatically into mature military cyber forces. IISS's 2021 cyber power assessment finds India's offensive capabilities nascent and lacking doctrinal integration (International Institute for Strategic Studies 2021), confirming that technological capacity is necessary but insufficient.

Strategic culture provides partial explanatory value. Germany's post-war political culture of military restraint aligns with a defensive posture, while American strategic culture emphasizes

forward engagement and power projection (Kier 1995) (Posen 1984). Yet culture alone cannot account for India's offensive turn despite its historic strategic restraint, nor Germany's refusal to shift doctrine after significant attacks. Cultural effects are mediated through, and overruled by, domestic institutional structures.

5.3 Interaction Effects: A Conditional Theory

Comparative evidence supports a conditional theory: cyber doctrine in democracies emerges from the interaction of threat environment, institutional structures, and alliance context, with institutional constraints operating as the dominant mediating variable.

1. Germany: Constitutional and judicial institutions constrain offensive doctrine regardless of threat severity or alliance pressure.

2. India: Weak institutions create no veto points, allowing the threat environment to drive offensive development directly, independent of alliances.

3. United States: Moderate constraints permit offense externally while shaping doctrinal implementation, producing offense under severe threat conditions.

Core finding: The interaction between H1 and H2 determines doctrinal outcomes. Threats matter, but only institutional structures decide whether threats are interpreted through an offensive or defensive lens.

6 Policy Implications

6.1 Alliance Coordination: Embrace Differentiation

The comparative findings show that institutional differences among democracies create structurally divergent cyber doctrines. Yet NATO continues to pursue doctrinal harmonization and common responses (North Atlantic Treaty Organization 2023), overlooking that members vary in their legal capacity to conduct offensive cyber operations. Germany's constitutional privacy protections, anchored in Article 10 of the Grundgesetz (Federal Republic of Germany 1949) and repeatedly

reinforced by the BVerfG (Federal Constitutional Court of Germany 2020), form binding legal constraints, not negotiable political preferences. No amount of alliance pressure can induce Germany to adopt U.S.-style defend-forward operations without constitutional revision.

Policy Implication 1: Adopt differentiated alliance roles. Instead of trying to achieve conformity, NATO should codify role differentiation based on domestic institutional capability. Countries that enjoy strong privacy laws and effective regulatory authorities (like Germany, France, and the Netherlands) can be well-suited to spearhead efforts to build resilience in areas such as protecting critical infrastructure, validating threat intelligence, incident response, and establishing standards. Less regulated countries (such as the US and the UK) are well-suited to spearheading tasks such as offense and preemption.

Policy Implication 2: Prioritize deterrence by denial over punishment. Deterrence through denial, hardening systems, and reducing the likelihood of successful attacks faces fewer institutional constraints than deterrence by punishment. Because punishment requires offensive cyber capabilities that many democracies cannot legally authorize, denial provides a more credible and widely adoptable strategy. Instead of punishing enemy attacks, NATO can maintain resilience investments, share responses among allies, attribute attacks to public exposure, and pursue diplomatic or economic responses that do not involve cyber-offensive escalations.

6.2 Engaging Emerging Democracies: Institutional Development First

India illustrates the challenges and opportunities of collaboration in emerging democratic nations whose cyber development agenda is fueled by threat environments that are very hostile and comprised of weakened institutional safeguards. Rapid advancements in offensive capabilities in such circumstances may be dangerous for diffusion and lead to instability.

Policy Implication 3: Build institutions before transferring advanced capabilities. Cyber assistance programs for partners such as India must first and foremost build institutional capacity: private international law regimes, semi-autonomous agencies, judicial review arrangements, and independent monitoring. Collaboration must be multilayered: capacity building for defense

first, and only more extensive intelligence sharing if capacity for offense is in place.

Policy Implication 4: Craft norms around outcomes rather than means. Global cyber norms should focus on shared outcomes, such as prohibiting peacetime attacks on civilian power infrastructure, while allowing states to achieve these outcomes through institutionally compatible means. This approach accommodates Germany’s defensive regulatory model, the U.S. offensive-enabled deterrence model, and India’s hybrid trajectory, producing norms that are more feasible and durable.

7 Conclusion

Why do democracies confronting similar cyber threats adopt divergent doctrines? This paper shows that variation stems from the interaction of three forces: threat environment, institutional constraints, and alliance structures, with domestic institutions exerting the most powerful independent influence. Threats matter, alliances matter, but neither can be understood without accounting for the domestic legal and organizational environment through which they are interpreted.

The United States developed an offense-dominant doctrine because of intense threats, permissive legal frameworks for foreign operations, and the demands of leadership within intelligence and security alliances. Germany, despite experiencing serious intrusions, remains defense-dominant because constitutional privacy protections, judicial oversight, and powerful civilian regulatory institutions constrain the state’s ability to adopt offensive capabilities. India, facing acute regional threats and operating under comparatively weak institutional constraints, has rapidly and autonomously developed offensive capabilities, with minimal shaping from alliances.

These findings challenge the assumption that democratic regimes are expected to unite on cyber doctrines. Rather, the observations above demonstrate that diversity in international systems leads to divergent cyber doctrines emerging and persisting despite shared enemies among states. Immediately, this implies that any state attempting to achieve its strategic aims in an alliance on the back of similar doctrinal positions will fail.

For emerging democracies, the findings highlight the need for both institutional and ca-

pability building. In the absence of strong safeguards and regulatory autonomy, cyber-offensive capacity may develop in ways that undermine democratic control. In the end, the meaning of cyber doctrine can only be comprehended within the understanding of democratic systems that govern the perception of threats, interpretation of obligations, and decisions between offense and defense that states must adhere to. Cyber strategy is more about reconciling security needs and defining principles of democratic systems than anything else.

References

- Alghamdi, Mohammed I. (2021). “Impact of Cyber Attack on Saudi Aramco”. In: *Journal of Cybersecurity and Information Management (or appropriate journal name)*. “Impact of Cyber Attack on Saudi Aramco (Shamoon 2012)”.
- Bennett, Andrew and Jeffrey T. Checkel (2015). *Process Tracing: From Metaphor to Analytic Tool*. Cambridge University Press.
- Bundesamt für Sicherheit in der Informationstechnik (2015). *The IT Security Situation in Germany 2015*. Federal Office for Information Security.
- Bundesamt für Sicherheit in der Informationstechnik (BSI) (2022). *BSI-Lagebericht zur IT-Sicherheit 2022*. Bundesamt für Sicherheit in der Informationstechnik.
- (2023). *Tasks and Functions of the BSI*.
- Bundesamt für Verfassungsschutz (BfV) (2022). *Verfassungsschutzbericht 2021*. Federal Ministry of the Interior and Community.
- Bundeswehr (2017). *Aufstellungsappell für das Kommando Cyber- und Informationsraum*.
- Cybersecurity and Infrastructure Security Agency (2025). *About CISA*.
- EBSCO Research Starters (2021). *Operation Ababil (2012)*. URL: <https://www.ebsco.com/research-starters/political-science/operation-ababil-2012>.
- European Union (2016a). *Directive (EU) 2016/1148 (NIS Directive)*. Official Journal of the European Union, L194.

- European Union (2016b). *Regulation (EU) 2016/679 (General Data Protection Regulation)*. Official Journal of the European Union, L119.
- Federal Bureau of Investigation (2014). *FBI Update on Sony Investigation*.
- Federal Constitutional Court of Germany (2008). *Judgment of 27 February 2008, 1 BvR 370/07*.
— (2020). *Judgment of 19 May 2020, 1 BvR 2835/17*.
- Federal Ministry of the Interior and Community (2021). *Cybersecurity Strategy for Germany 2021*.
- Federal Republic of Germany (1949). *Basic Law for the Federal Republic of Germany (Grundgesetz), Articles 10 and 13*.
- George, Alexander L. and Andrew Bennett (2005). *Case Studies and Theory Development in the Social Sciences*. MIT Press.
- Gioe, David, Michael Goodman, and David Schaefer (2021). *Five Eyes: the past, present and future of the world's key intelligence alliance*. URL: <https://engelsbergideas.com/essays/five-eyes-the-past-present-and-future-of-the-worlds-key-intelligence-alliance/>.
- Government of India (2023). *Digital Personal Data Protection Act, 2023*. Act No. 22 of 2023.
- Indian Computer Emergency Response Team (CERT-In) (2004). *Indian Computer Emergency Response Team (CERT-In)*. URL: https://www.digitalindia.gov.in/di_ecosystem/indian-computer-emergency-response-team-icert/.
— (2021). *Annual Reports 2016–2021*.
— (2022a). *Annual Report 2022*.
— (2022b). *Directions under Section 70B of the IT Act*.
- International Institute for Strategic Studies (2021). *Cyber Capabilities and National Power: A Net Assessment*. IISS.
- Katzenstein, Peter J. (2003). “September 11 in Comparative Perspective”. In: *International Organization* 57.4, pp. 731–760.
- Kier, Elizabeth (1995). “Culture and Military Doctrine”. In: *International Security* 19.4, pp. 65–93. DOI: 10.2307/2539120.

- Kostyuk, Nadiya and Yuri M. Zhukov (2019). “Invisible Digital Front”. In: *Journal of Conflict Resolution* 63.2, pp. 317–347.
- Libicki, Martin C. (2009). *Cyberdeterrence and Cyberwar*. RAND Corporation.
- Lindsay, Jon R. (2014). “The Impact of China on Cybersecurity”. In: *International Security* 39.3, pp. 7–47.
- Ministry of Electronics and Information Technology (MeitY) (2021). *Data Protection Framework: Status Update*.
- Ministry of External Affairs (MEA) (2018). *India–Israel Joint Statement: Cyber Cooperation*.
- Nakashima, Ellen (2018). *U.S. Cyber Command Operation Protected the 2018 Elections*. URL: <https://www.washingtonpost.com/world/national-security/us-cyber-command-operation-protected-midterms/2018/10/24/>.
- NASSCOM (2023). *Indian Tech Industry: Workforce and Growth Statistics 2023*.
- National Critical Information Infrastructure Protection Centre (NCIIPC) (2014). *NCIIPC Guidelines and Mandate*.
- North Atlantic Treaty Organization (2016a). *Cyber Defence Pledge*.
- (2016b). *Warsaw Summit Communiqué*.
- (2023). *Cyber Defence*.
- NSA, CISA, FBI (2021). *Russian SVR Cyber Actors Targeting U.S. and Allied Networks*.
- Nye, Joseph S. (2017). “Deterrence and Dissuasion in Cyberspace”. In: *International Security* 41.3, pp. 44–71. DOI: 10.1162/ISEC_a_00266.
- Posen, Barry R. (1984). *The Sources of Military Doctrine*. Cornell University Press.
- Press Information Bureau, Government of India (2021). *Establishment of the Defence Cyber Agency*. URL: <https://www.pib.gov.in/PressReleasePage.aspx?PRID=1742322>.
- Quad Leaders (2021). *Joint Statement from Quad Leaders*.
- Recorded Future Insikt Group (2021). *RedEcho Targeting Indian Power Sector*. URL: <https://www.recordedfuture.com/research/redecho-targeting-indian-power-sector>.

Reuters (2014). *Las Vegas Sands Network Hit by Destructive Malware*. URL: <https://www.reuters.com/article/technology/las-vegas-sands-network-hit-by-destructive-malware-in-feb-bloomberg-idUSKBN0JQ045/>.

Sanger, David E. (2016). *U.S. Cyberattacks Target ISIS in a New Line of Combat*. URL: <https://www.nytimes.com/2016/04/25/us/politics/us-directs-cyberweapons-at-isis-for-first-time.html>.

Supreme Court Observer (2017). *Puttaswamy v. Union of India*.

The White House (2023). *National Cybersecurity Strategy*.

U.S. Congress (2018). *John S. McCain National Defense Authorization Act for Fiscal Year 2019*. Public Law 115–232.

U.S. Department of Defense (2023). *2023 DoD Cyber Strategy Summary*. U.S. Department of Defense.

U.S. House Committee on Oversight and Government Reform (2016). *The OPM Data Breach*.

United States Congress (1978). *Foreign Intelligence Surveillance Act of 1978*. Public Law 95–511.

— (2015). *USA FREEDOM Act of 2015*. Public Law 114–23.

Van Evera, Stephen (1999). *Causes of War*. Cornell University Press.